

Data Driven Business Strategy

Capstone Project Report



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Executive Summary

The Business

ChargePoint is a leading electric vehicle (EV) charging network provider in the United States, offering a comprehensive ecosystem of charging hardware, software, and services. The company operates one of the largest networks of independently owned EV charging stations, connecting drivers to a wide range of public and private charging locations.

ChargePoint's business model focuses on providing charging solutions to commercial businesses, fleet operators, residential complexes, and municipalities. In addition to selling charging hardware, the company generates recurring revenue through subscription-based software services that enable station owners to manage operations, monitor usage, and optimize performance.

As EV adoption continues to accelerate across the United States, ChargePoint faces the strategic challenge of expanding its charging infrastructure efficiently. Identifying high-demand locations and ensuring optimal station utilization are critical to maximizing return on investment and maintaining a competitive advantage in the rapidly growing EV market. This project aims to support ChargePoint's infrastructure expansion strategy by leveraging data analytics to identify demand-supply mismatches and recommend optimal locations for future charging station deployment.

Business Concept

This project focuses on developing a data-driven strategy for optimizing EV charging infrastructure expansion. By analyzing EV adoption (demand) and charging station availability (supply), the project aims to identify regions where infrastructure investment can maximize utilization and return on investment. The goal is to support strategic decision-making for ChargePoint in expanding its network efficiently. This approach is supported by the integration of EV population data (demand) and charging station availability data (supply).

Business Problem / Opportunity

As electric vehicle adoption continues to grow rapidly across the United States, charging infrastructure expansion has not kept pace with demand. This creates a mismatch where some regions experience high congestion and unmet demand, while others have underutilized charging stations.

For ChargePoint, inefficient infrastructure placement can result in lower utilization rates, reduced revenue, and higher capital risk. At the same time, this mismatch presents a significant opportunity to strategically expand charging infrastructure in high-demand, underserved areas.

Objective & Goals

The primary objective of this project is to identify optimal locations for EV charging infrastructure expansion using data-driven analysis. The key goals include:

- Identifying regions with high EV demand and limited charging infrastructure
- Analyzing utilization patterns of existing charging stations
- Estimating potential return on investment (ROI) for infrastructure expansion
- Providing strategic recommendations for ChargePoint's growth and market positioning

Product / Service Overview

ChargePoint provides EV charging hardware, software, and network services that enable businesses, property owners, and fleet operators to install and manage charging stations. The company's platform allows users to locate charging stations, monitor usage, and manage charging operations efficiently. This project focuses specifically on ChargePoint's infrastructure expansion services and network optimization capabilities.

Target Market

ChargePoint primarily serves commercial businesses, fleet operators, residential complexes, and municipalities. The most relevant target segments for this project are regions with rapidly growing EV adoption and insufficient charging infrastructure, as these represent the highest potential for expansion and revenue generation.

Unique Value Proposition

ChargePoint differentiates itself by offering an integrated solution that combines hardware, software, and network services. Unlike competitors that focus primarily on increasing the number of charging stations, ChargePoint has the opportunity to leverage data analytics to optimize infrastructure placement. This project enhances that value by providing a data-driven approach to identifying high-demand locations, improving utilization rates, and maximizing return on investment.

This project enables ChargePoint to transition from general infrastructure expansion to a targeted, data-driven investment strategy, improving both utilization efficiency and long-term profitability.

Market Analysis & Strategy

Industry Trends: EV Growth and Charging Gap

The U.S. electric vehicle (EV) market has grown rapidly over the past few years, showing a clear shift toward electrification. EV sales reached approximately 1.4 to 1.6 million units in 2024, accounting for about 10% of total vehicle sales. While this indicates strong growth, it also shows that the U.S. is still behind the global average, where EV penetration has already exceeded 20%. This suggests that there is still significant room for growth in the U.S. market, which will likely increase the demand for supporting infrastructure in the near future.

At the same time, charging infrastructure has been expanding, but not at the same pace as EV adoption. As shown in Figure 1, the total number of public charging ports has steadily increased from 2019 to 2023. However, most of this growth comes from Level 2 chargers, while DC fast chargers, which are more critical for long-distance travel and high-usage scenarios, remain in a relatively small portion of the total. This indicates that although the number of chargers is increasing, the quality and efficiency of the infrastructure may not fully match user needs.

More importantly, the growth pattern shown in Figure 1 suggests that infrastructure expansion is relatively stable and linear, while EV adoption is accelerating more quickly. This creates a growing gap between demand and supply. In other words, even though there are more chargers being built, they may not be enough to support the increasing number of EV users. This mismatch is a key issue in the industry and forms the foundation of our project, which focuses on identifying where infrastructure expansion is most needed.

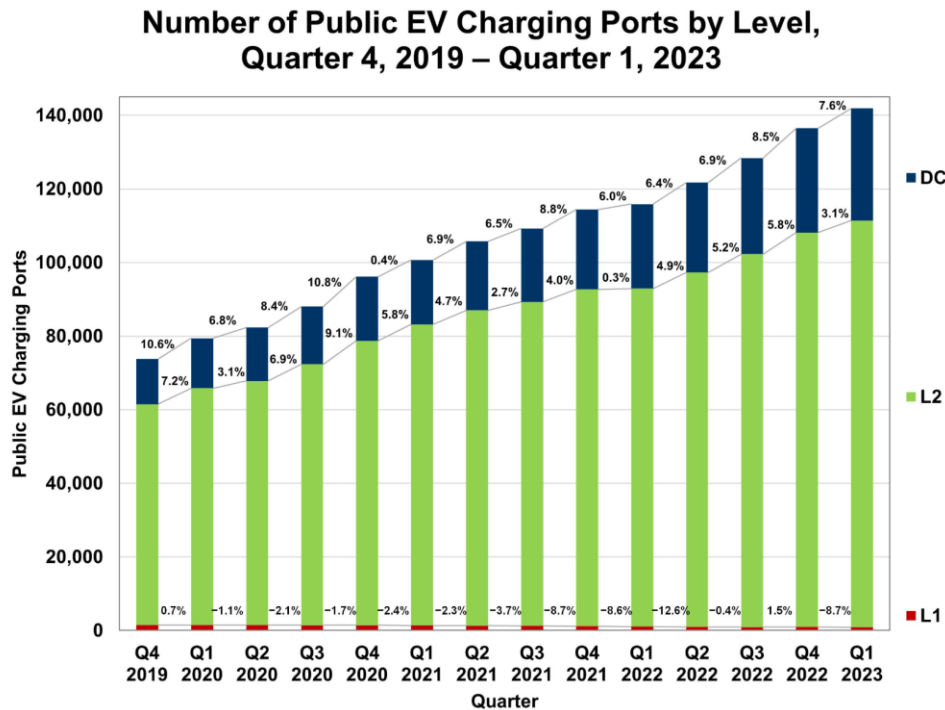


Figure 1. Number of public EV charging ports by level (Level 1, Level 2, and DC fast chargers) in the United States, Q4 2019 – Q1 2023.

Geographic Distribution and Demand Imbalance

In addition to the overall growth gap, another important issue in the EV market is the uneven geographic distribution of charging infrastructure. As shown in Figure 2, charging stations are heavily concentrated in a small number of states, particularly California, Texas, and New York. These states have significantly higher numbers of charging units compared to others, while many states have relatively limited infrastructure.

This uneven distribution suggests that access to charging infrastructure varies significantly depending on location. In high-density states, users may have better access to charging, but they may also experience congestion due to high demand. In contrast, in less-developed regions, the lack of charging infrastructure can discourage EV adoption altogether. This creates a double problem: some areas may have too much demand for the available infrastructure, while others may have underdeveloped markets due to limited access.

From a data perspective, this imbalance can be further analyzed by comparing EV population data with charging station data across regions. For example, calculating the ratio of EVs to charging stations in each state can help identify areas where infrastructure is

insufficient relative to demand. This type of analysis directly supports our research question, which focuses on identifying regions with the highest demand–supply mismatch.

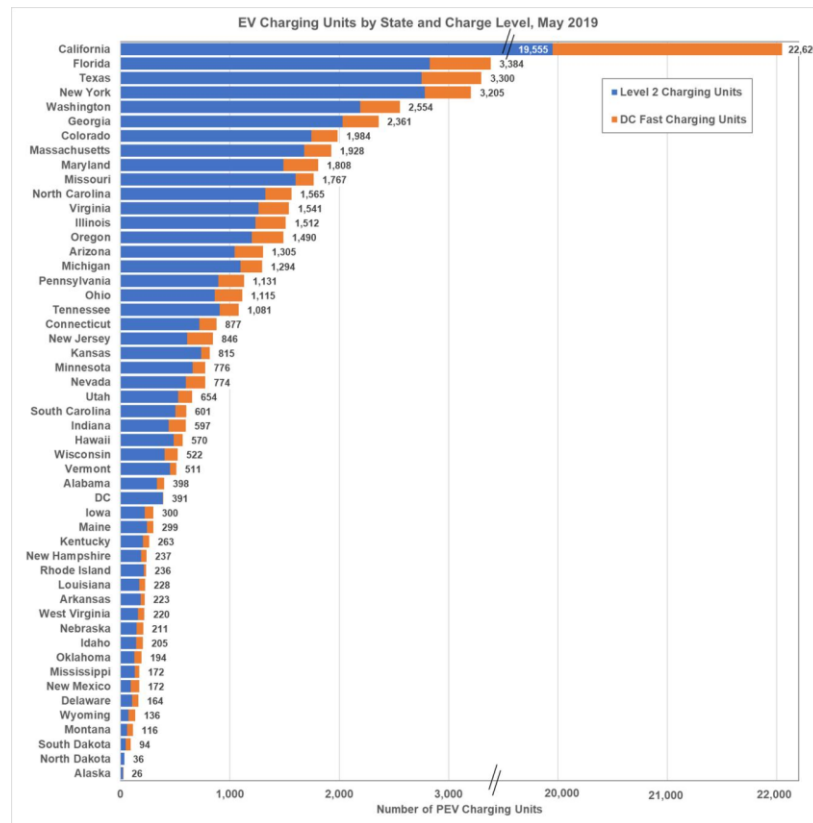


Figure 2. EV charging units by state in the U.S.

Target Customers

The EV charging market serves several key customer segments, including commercial property owners, fleet operators, residential complexes, and local governments. Each of these groups has different needs, but they all depend on reliable and accessible charging infrastructure.

Fleet operators are one of the most sensitive customer groups because their operations depend heavily on charging efficiency. If charging stations are limited or poorly located, it can directly impact vehicle availability and operational costs. Similarly, commercial property owners and residential complexes may install charging stations to attract customers or tenants, especially in areas where EV adoption is increasing. For these customers, the availability of charging infrastructure can be a key factor in competitiveness and customer satisfaction.

Based on the trends shown in Figures 1 and 2, the most valuable target customers are likely located in regions where EV adoption is growing quickly but infrastructure is still

limited. These customers have the strongest need for additional charging solutions, making them ideal targets for expansion. Therefore, identifying these high-demand, underserved areas is essential not only for market analysis but also for developing an effective customer targeting strategy.

Competitor Overview

The EV charging market in the U.S. is becoming increasingly competitive as more companies invest in infrastructure. Major players include ChargePoint, EVgo, and Tesla's Supercharger network. Tesla has a strong competitive advantage due to its integrated ecosystem, where both vehicles and charging infrastructure are controlled by the same company. This allows Tesla to provide a more seamless user experience.

Based on the industry trends shown in the figures, however, most competitors focus primarily on increasing the number of charging stations rather than optimizing their locations. This approach can lead to inefficiencies, such as building stations in areas with low demand or failing to provide enough capacity in high-demand regions. As a result, even though the total number of charging stations is increasing, the overall system may still not operate efficiently.

This creates an opportunity for differentiation. Instead of focusing only on expansion, companies like ChargePoint can use data analytics to better understand demand patterns and improve location decisions. By identifying where infrastructure is most needed, the company can achieve higher utilization rates and better returns on investment compared to competitors who follow a less targeted expansion strategy.

SWOT Analysis

From a strategic perspective, ChargePoint has several strengths that position it well in the EV charging market. Its large network and combination of hardware and software services allow it to offer a comprehensive solution to customers. The subscription-based model also provides a more stable and predictable revenue stream compared to one-time hardware sales.

However, the company also faces challenges. One major issue is the high cost of infrastructure deployment, especially if charging stations are not used efficiently. As shown in the earlier figures, uneven distribution and mismatch between supply and demand can lead to underutilization in some areas, which reduces overall profitability.

At the same time, the market presents strong opportunities. The continued growth of EV adoption increases the need for charging infrastructure, especially in underserved regions. The mismatch identified in the data suggests that there are clear areas where

additional investment could generate high returns. On the other hand, threats include increasing competition, policy uncertainty, and the risk of overexpansion in low-demand areas.

Connection to Company Strategy

Based on the analysis above, the key issue in the EV charging market is not simply the total number of charging stations, but how effectively they are distributed. The figures show that EV adoption is growing faster than infrastructure, and that infrastructure is unevenly distributed across regions. This indicates that there are specific areas where demand is high, but supply is limited.

For ChargePoint, this suggests that future growth should focus on targeted expansion rather than general expansion. By using data such as EV population, charging station locations, and usage patterns, the company can identify regions with the highest demand–supply mismatch. These regions represent the best opportunities for new infrastructure investment.

In addition, focusing on high-demand areas can improve utilization rates and increase return on investment (ROI). Instead of spreading resources evenly across all regions, a more data-driven approach allows the company to prioritize locations where charging stations are most likely to be used. This not only improves operational efficiency but also strengthens the company’s competitive position in the market.

Data & Descriptive Analysis

This presents the dataset overview, key variables, data cleaning procedures, summary statistics, and descriptive trends underpinning the infrastructure expansion analysis. All statistics were derived directly from the two source datasets as described below using Python and Power Bi,

Dataset Overview

This analysis combines two (2) publicly available datasets to study the alignment between electric vehicle (EV) adoption and public charging infrastructure across Washington State. Both datasets were cleaned, standardized, and merged at the city level to enable comparison of EV demand against charging supply.

Dataset 1 - EV Population

Field	Value
Source	Washington State Department of Licensing (DOL)
Dataset name	Electric Vehicle Population Data
Date accessed	April 2026
Raw record count	280,833 rows (one per registered vehicle)
WA State records	2800,131 (702 out-of-state records removed)
Geographic scope	Washington State (all 39 counties)
Vehicle types covered	Battery Electric Vehicle (BEV) & Plug-in Hybrid EV (PHEV)
Model years present	2018 - 2026
Number of columns	16 original fields

EV Population – Column Reference

Column Name	Data Type	Description
VIN (1-10)	Text	First 10 characters of the Vehicle Identification Number (VIN).
County	Text	Geographic region of the registered vehicle owner.
City	Text	City where the registered vehicle owner resides.
Postal Code	Text	5-digit ZIP code for the vehicle owner's residence.
Model Year	Text	Vehicle model year decoded from VIN.
Make	Text	Vehicle manufacturer (e.g., Tesla, Nissan)
Model	Text	Vehicle model name.
Electric Vehicle Type	Text	BEV (Battery Electric) or PHEV (Plug-in Hybrid).
CAFV Eligibility	Text	Indicates whether the vehicle qualifies as a Clean Alternative Fuel Vehicle under WA law.
Electric Range	Number	Distance the vehicle can travel using only electric power (EPA-rated).
Legislative District	Text	WA State legislative district number.
DOL Vehicle ID	Text	Unique identifier assigned by the Department of Licensing.
Vehicle Location	Point	Geographic coordinates (lat/lon) represent the center of the ZIP code.

Electric Utility	Text	Utility provider(s) serving the vehicle's registration address.
2020 Census Tract	Text	Census tract identifier (GEOID) from the 2020 U.S. Census.

Dataset 2 - Alternative Fuel Stations

Field	Value
Source	U.S. Department of Energy - National Renewable Energy Laboratory (NREL)
Dataset name	Alternative Fuel Station Locator
Date accessed	April 2026
Raw record count	79,566 rows (all fuel types, U.S. states)
WA electric stations (after filter)	3,009 records
Geographic scope	WA State (Fuel Type = ELEC, State = WA)
Access type	100% Public access stations after filtering
Number of original columns	75 columns (reduced to 15 after cleaning)

Fuel Stations – Key Column Reference

Column Name	Data Type	Description
Station Name	Text	Name of the charging station or host business
City/State/ZIP	Text	Geographic location – City is the primary join key to EV dataset
Station Phone	Text	The phone number of the station
Status Code	Text	Current station status: E available, P planned, T temporarily unavailable.
Groups With Access Code	Text	Legacy field describing who can access the station.
Access Days Time	Text	Hours of operation.
Cards Accepted	Text	Payment methods accepted at the station.
EV Level1 EVSE Num	Integer	Number of Level 1 EV charging ports.
EV Level2 EVSE Num	Integer	Number of Level 2 EV charging ports.
EV DC Fast Count	Integer	Number of DC fast charging ports.

EV Network	Text	Charging network names, such as ChargePoint, EVgo, Tesla, or Non-Networked.
EV Network Web	Text	Website of the EV charging network.
Geocode Status	Text	Accuracy level of the latitude and longitude.
Latitude	Decimal	Latitude coordinate of the station.
Longitude	Decimal	Longitude coordinate of the station.
Date Last Confirmed	Date	Date, the station information was last confirmed.
ID	Integer	Unique station identifier.
Updated At	DateTime	Date and time the station record was last updated.
Owner Type Code	Text	Type of station owner, such as private, municipal, state, federal, or utility.
Federal Agency ID	Integer	If the station is owned by a federal agency, a unique identifier for the federal agency that owns this station.
Federal Agency Name	String	If the station is owned by a federal agency, The name of the federal agency that owns this station.
Open Date	Date	Date the station began offering charging service.
EV Connector Types	Text/Array	Connector types available, such as J1772, Tesla/NACS, CCS, or CHAdeMO.
Access Code	Text	Indicates whether the station is public or private.
Access Detail Code	Text	Extra access conditions, such as call ahead, limited hours, or credit card required.
Facility Type	Text	Type of location where the charger is installed, such as airport, hotel, parking garage, or workplace.
EV Pricing	Text	Pricing information for using the charging station.
EV On-site Renewable Source	Text	Renewable energy sources used on-site, if available, such as solar or wind.
Restricted Access	Boolean	Indicates whether a public station is restricted to patrons, guests, or customers.
Maximum Vehicle Class	Text	Maximum vehicle size that can physically access the station: LD, MD, or HD.

EV Workplace Charging	Boolean	Indicates whether the station is mainly used for employee/workplace charging.
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Key Variables

The analysis uses primary variable groups that collectively describe the demand-supply relationship between EV adoption and charging infrastructure at the city level.

Demand Variables

- *EV_Registrations*: Total registered EVs per city (BEV + PHEV) represents charging demand. Range: 1 – 43,219. Highly skewed as Seattle accounts for 15.4% of statewide registrations.
- *Electric Vehicle Type*: BEV (80.2% => 224,606 vehicles) vs PHEV (19.8% => 55,525 vehicles). BEVs depend entirely on charging infrastructure; PHEV can use gasoline as backup. BEV-only analysis provides a more conservative demand estimate.

Supply Variables

- *Station_Count*: Number of physical charging station locations per city. Washington has 3,009 public stations.
- *Level2_Ports*: Sum of Level 2 EVSE ports per city; WA total: 6,313 ports. Slower charge speed (app. 4-8 hrs. for full charge) -most common type (75% of all ports),
- *DCFast_Ports*: Sum of DC Fast charging ports per city; WA total: 2,091 ports. Charge speed of (app. 20-45min). Only 16% of WA stations have DC Fast capability.
- *Total_Ports*: Level2_Ports + DCFast_Ports, WA statewide total: 8,404 ports; primary supply metric used throughout the analysis.

Gap and Utilization Variables

- *EVs_per_Port (Gap Score)*: $EV_Registration \div (Total_Ports + 1)$ => Core gap metric. Higher value = more EVs sharing each port = higher infrastructure strain, A threshold of 50 was used to classify “Expansion Needed” cities. Statewide average = 33.3,
- *Category*: A 3-level classification was used as the primary segmentation variable across all Power BI pages: (1) No Infrastructure=>Total_Ports =0; Expansion Needed =>EVs_per_port > 50; (3) Adequately Served => all others.

Data Cleaning

Both datasets required cleaning before analysis. All cleaning steps were performed in Power BI Desktop and Python. The steps below are documented in order of application using Power BI Desktop.

EV Population Dataset

	Step	Action	Reason
1	Filter State = WA	Removed 702 out-of-state records (CA, VA, FL, MD, TX, OR, CO, AZ, IL)	Analysis scope is Washington State only
2	Remove null City rows	12 rows removed where City, County, Postal Code were null	Cannot join to infrastructure data without a city name
3	Filter Model Year	Kept 2018-2016 model years only	Focus on the current EV generation removes 1999-2017.
4	Standardize City casing	Applied Text.Proper (Capitalize Each Word) to City and County columns	Ensures consistent join key when merging with Fuel Stations
5	Convert Postal Code to Text	Changed data type from Number to Text	Preserves leading zeros in ZIP codes
6	Replace null	Replaced 703 nulls with 0	Prevents aggregation errors; column not used in core analysis

Fuel Stations Dataset

	Step	Action	Reason
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1	Filter Fuel Type = ELEC	Retained only electric charging stations	Analysis is specific to EV charging infrastructure
2	Filter State = WA	Reduced from 79,566 national records to 3,009 WA records	Matches geographic scope of EV population data
3	Remove null City rows	Removed 3 rows with no City value	Cannot participate in city-level join without a city name
4	Replace null Level2 ports	Removed 390 nulls with 0 in EV Level2 EVSE Num	Ensures SUM counts correctly as nulls would cause undercounting
5	Replace null DC Fast ports	Replaced 2,528 nulls with 0 in EV DC Fast Count	Same reason as Level 2 with 84% of records with null DC Fast count
6	Standardize City casing	Applied T.Proper to City column	Must match EV Population City casing for joining to work
7	Remove unused column	Reduced from 75 to 15 key columns	Improves Power BI model performance and reduces file size
8	Convert ZIP to Text	Changed data type from Number to Text	Preserves leading zeros in ZIP codes

Merge and Calculated Columns

	Step	Detail
1	Group EV by City	EV_Population grouped by City --> EV_Registrations (CountRows)
2	Group Fuel by City	Fuel_Stations grouped by City ---> Station_Count, Level2_Ports (Sum), DCFast_Ports (Sum)
3	Left Outer Merge	EV_City merged with Fuel_City on City --> Left Outer Join preserves all 554 EV cities including 313 with zero stations
4	Total_Ports column	Custom column: Level2_Ports + DCFast_Ports
5	EVs_per_Port column	Customn column: EV_Registrations ÷ (Total_Ports + 1)
6	Category column	If Total_Ports = 0 then “No Infrastructure” else if EVs_per_Port > 50 then “Expansion Needed” else “Adequately Served”

Analytical Table – Merge and Calculated Columns (WA_EV_Charging_Gap)

An analytical table was derived by aggregating both source datasets at the city level and merging on City name using a Left Outer Join. This table is the foundation for all Power BI visuals, DAX measures, and research questions analysis.

Column	Definition
City	Join key; standardized to Title Case in both datasets before merging
County	Carried forward from EV_Population after grouping
EV_Registrations	COUNT of EV records per city from EV Population dataset
Station_Count	COUNT of station records per city from Fuel Stations dataset
Level2_Ports	SUM of EV Level2 EVSE NUM per city (nulls --> 0 before summing)
DCFast_Ports	SUM of EV DC Fast Count per city (null --> 0 before summing)
Total_Ports	Level2_Ports + DCFast_Ports --> total charging capacity per city
EVs_per_Port	$EV_Registrations \div (Total_Ports + 1)$ --> demand-supply strain ratio
Gap_Score	Same as EVs_per_Port ---> primary ranking metric for Priority analysis
Category	No Infrastructure / Expansion Needed / Adequately Served

Summary Statistics

The following statistics are verified directly against the KPI card values and the WA_EV_Charging_Gap analytical table and are represented as summary visuals on the Power BI dashboard.

Statewide Overview – KPI Card Values

Metric	Value
Total EVs (COUNTROW of EV_Population)	280,131
Total Ports (SUM of WA_EV_Charging_Gap[Total_Ports])	9,342
Total Charging Stations	3,009 public stations
Cities With No Infrastructure	313 cities
Priorities Cities for Expansion	54 cities
Average EVs Sharing Each Port (AVERAGEX per city)	51.9
Statewide EVs ratio (Total EVs ÷ Total	33.3 EVs sharing each public port on average

EV Vehicle Type Breakdown

Vehicle Type	Count
Battery Electric Vehicle (BEV)	224,606 ---> 80.2% of total
Plug-in Hybrid Electric Vehicle (PHEV)	55,525 ---> 19.8% of total
Total WA EV Registrations	280,131

Charging Infrastructure Breakdown

Metric	Value
Total charging stations (WA)	3,009 public stations
Total Level 2 ports (raw fuel table)	6,313
Total DC Fast ports (raw fuel table)	2,091
Total ports	8,404
Total_Ports in WA_EV_Charging_Gap (Power BI KPI)	9,342
Average Level 2 ports per station	2.4 ports
Average DC Fast ports per station	4.3 ports
Stations with Level 2 only	2,528 (84%)
Stations with DC Fast capability	481(61%)
Top network - ChargePoint	1,826 stations (61%)
Second Network - Blink	298 stations (10%)
Tesla stations (Supercharger + Destination)	210 statiions (7%)
FLO network	147 stations (5%)
Non-networked stations	163 stations (5%)

Infrastructure Category Breakdown

Category	Cities (Count & %)
No Infrastructure (Total_Ports = 0)	313 cities – 56.5%
Expansion Needed (EVs_per_Port > 50)	54 cities – 9.7%
Adequately Served	187 cities – 33.8%
TOTAL Cities Analyzed	554 cities – 100%

Summary Statistics of Key Variables

	ev_count	ev_level2_evse_num	ev_dc_fast_count	total_ports	ev_per_port	gap_score
count	499.000000	499.000000	499.000000	499.000000	213.000000	499.000000
mean	561.342685	14.322645	4.268537	18.591182	61.765625	56.374961
std	2358.593646	95.635846	12.861325	104.917836	132.757954	124.523162
min	1.000000	0.000000	0.000000	0.000000	0.153846	0.142857
25%	6.000000	0.000000	0.000000	0.000000	7.872727	4.000000
50%	36.000000	0.000000	0.000000	0.000000	22.500000	13.750000
75%	269.000000	4.000000	1.000000	5.000000	63.611111	49.000000
max	43218.000000	1831.000000	156.000000	1987.000000	1214.000000	1080.000000

Trends and Visualizations

This section describes the key trends in the data and connects each trend to the corresponding Power BI and Python visualization and research question it addresses.

Rapid EV Adoption Growth Trend

EV registration in Washington grew rapidly between 2018 and 2023, reaching a peak of 6,0167 new registration in 2023 – a 102% increase over 2022. Growth moderated in 2024 and 2025, but cumulative adoption continues to increase far ahead of infrastructure expansion.

Model Year	New WA Registrations
2018	13,920
2019	10,754
2020	12,003
2021	20,674
2022	29,821
2023	60,167 <-- peak year

2024	49,406
2025	36,820
2026	19,667 (partial year – April 2026)

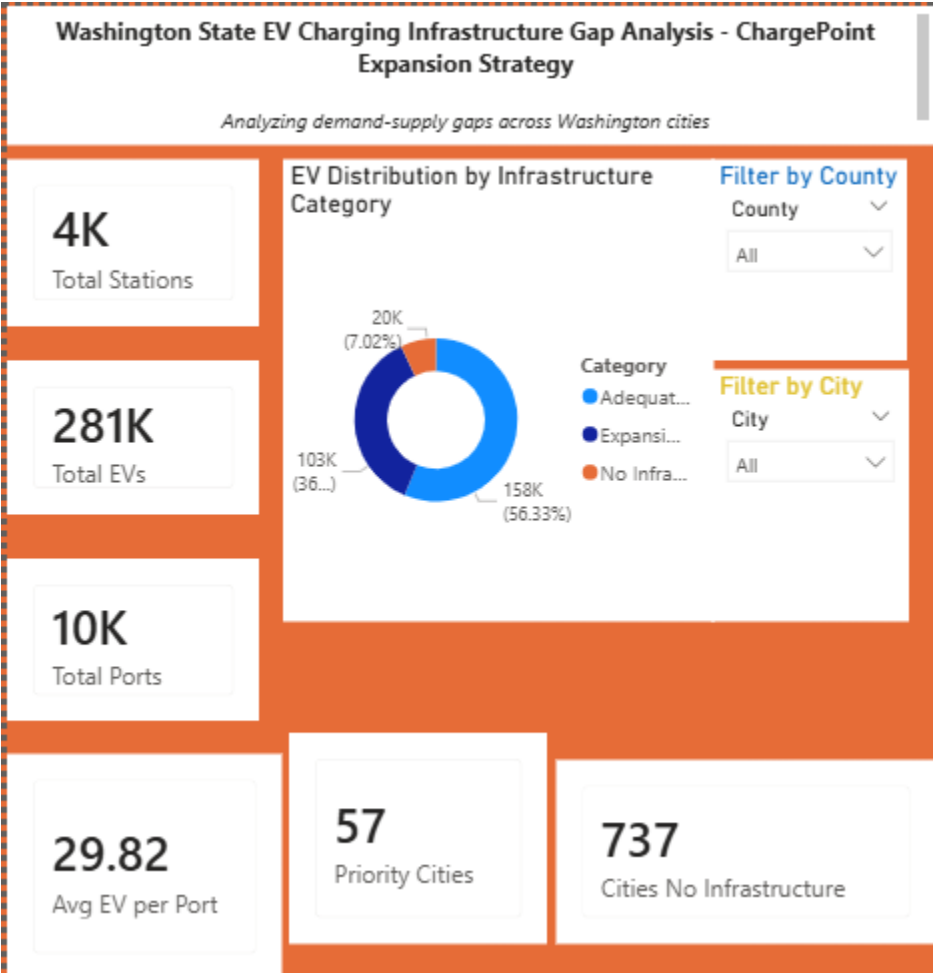


Figure 3: KPI cards Page

Power BI display- Total EV Registrations (280,131), Total Charging Ports (9,342), Total Stations (3,009), Cities with No Infrastructure (313), and Priority Cities for Expansion (54). The Avg EVs per Port card (51.9) quantifies the statewide infrastructure strain level for stakeholders

Geographic Concentration Trend

EV adoption is concentrated in King County (138,276 registrations – 49.4% of statewide total) followed by Snohomish (35,137 – 12.5%) and Pierce (23,041 – 8.2%). The top 3 counties account for 70.1% of all WA EV registrations. Charging infrastructure shows similar concentration – Seattle alone has 712 of the 3,009 stations (23.7%), but the suburban ring around Seattle had almost none.

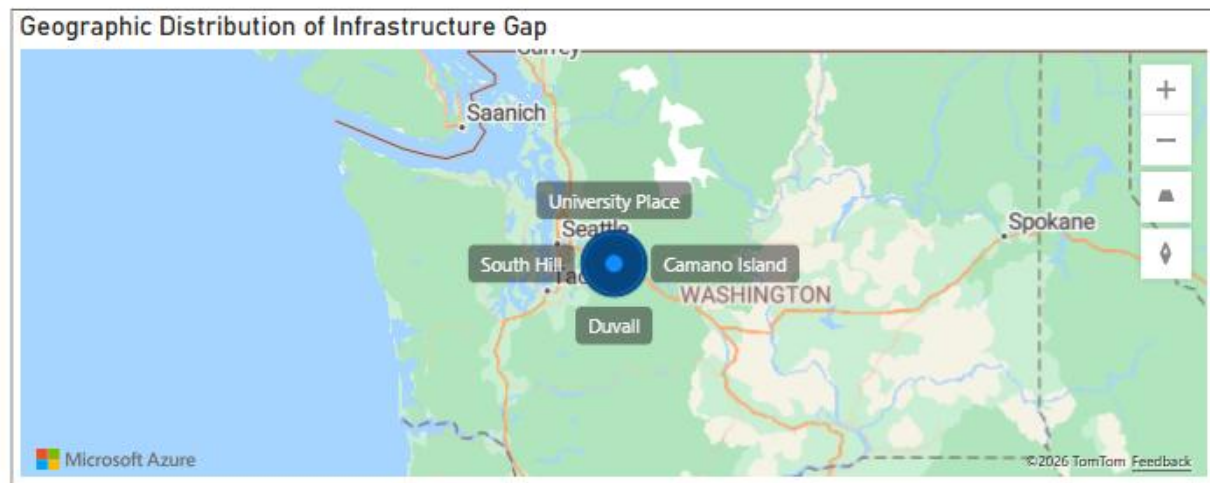


Fig 4: Map (Geographic Distribution)

The map on the Q1 page shows each city filtered to 'No Infrastructure' only, with bubble size proportional to EV_Registrations. This reveals the core spatial pattern: high-EV cities in the suburban Seattle fringe with no charging access at all. Cities like University Place, Duvall, and South Hill are visually prominent because of their large bubble size despite having zero ports.

Infrastructure Has Not Kept Up with Demand Trend

While registrations grew over 330% between 2010 and 2023, public charging infrastructure has not expanded proportionally. The statewide ratio of 30.0 EVs per port (total EVs / total ports) and the city-level leverage of 51.9 EVs per port both confirm significant infrastructure strain, 313 cities – 55.5% of all WA cities with EVs – have zero

public charging ports.

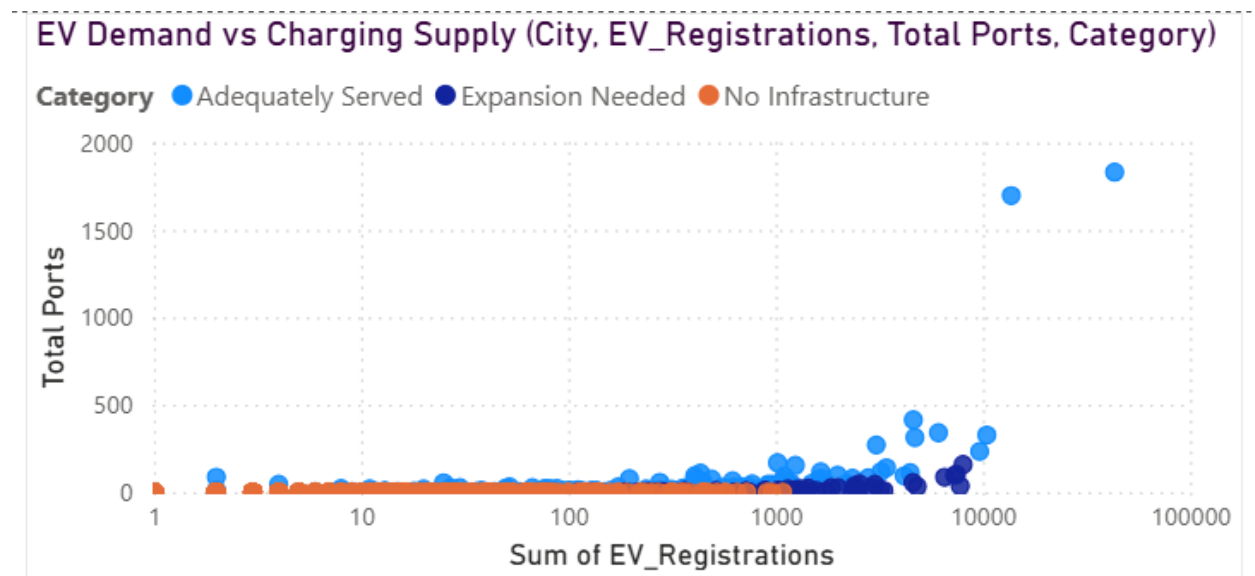


Fig 5: Scatter Chart

The EV Demand vs. Charging Supply scatter chart plots each city with EV_Registrations on the X-axis and Total_Ports on the Y-axis, with dots colored by Category. Cities above the trend line are underserved. “No Infrastructure” cities form a flat horizontal band near the X-axis (zero ports regardless of EV count). “Expansion Needed” cities cluster in the mid-range above the trend line. This addresses our Research question 2; how infrastructure utilization varies across regions.

Suburban Cities Face the Highest Utilization Strain Trend

Cities with the highest EVs_per_Port ratios are predominantly suburban areas in the Seattle metro; Mukilteo (607), Maple Valley (493), Lake Forest Park (491), Snohomish (303), and Sammamish (216). These cities share high household incomes, single-family home ownership enabling home charging, and tech-sector employment driving EV adoption, but public charging infrastructure has not followed residential growth patterns.

Top Cities with EVs But No Charging Infrastructure

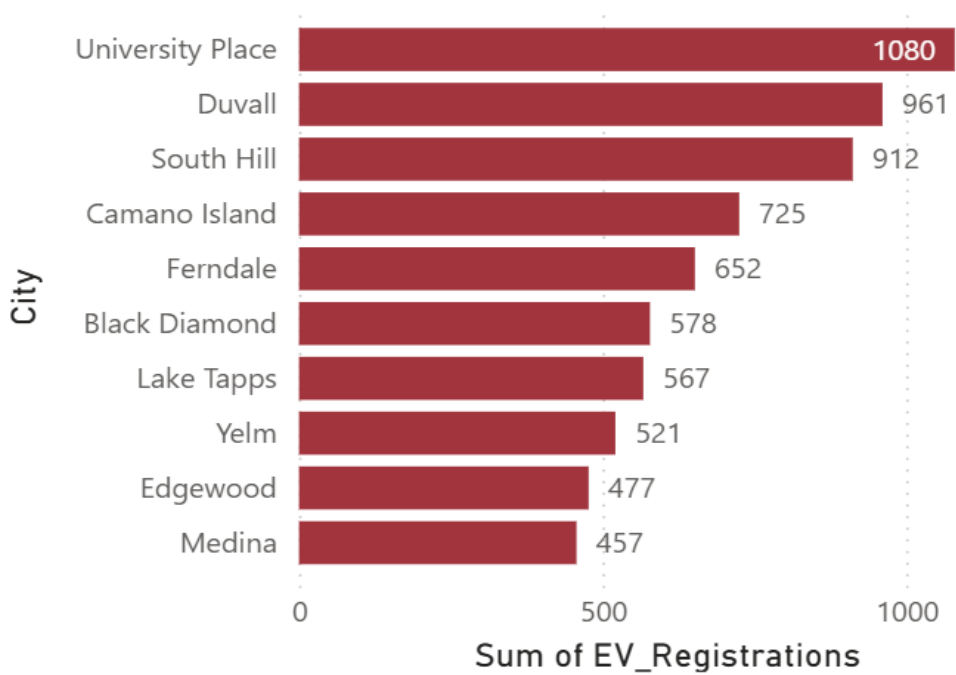


Fig 6: Column Chart

The Average Infrastructure Utilization by Category column chart compares the mean EVs_per_Port across the three categories with data labels showing exact values. 'Expansion Needed' cities show dramatically higher average utilization than 'Adequately Served' cities, quantifying the operational strain gap in a format accessible to non-technical stakeholders.

Priority Expansion Targets Are Concentrated and Actionable Trend

The 54 'Expansion Needed' cities already have proven charging station sites (permits, grid connections, host agreements), making them faster and lower-risk targets for ChargePoint deployment compared to the 313 cities that require full site development from scratch. Adding even 5–10 new ports in cities like Mukilteo or Maple Valley would reduce the EVs_per_Port ratio from 607 to under 100; an immediate, measurable impact.

Geographic Priority Map for Infrastructure Investment



Top Priority Cities for Expansion (City, Gap Score)

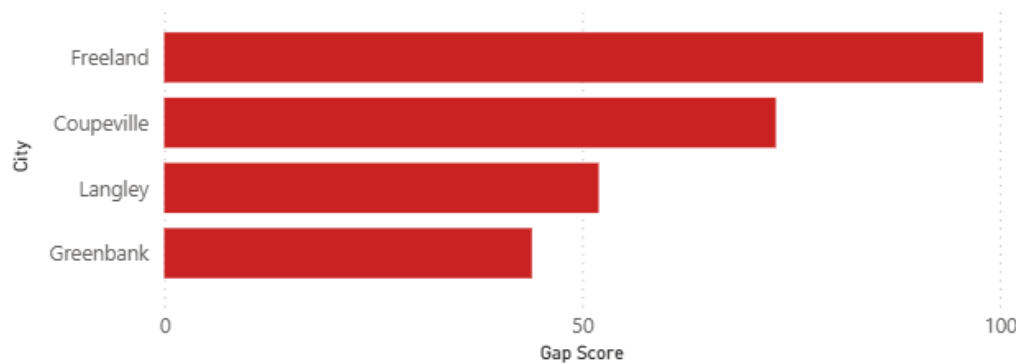


Fig 7: Bar Chart & Priority Map

The Top 15 Priority Cities bar chart ranks 'Expansion Needed' cities by Gap Score in descending order. The accompanying map shows these cities geographically with bubble size proportional to Gap Score and color saturation proportional to EVs_per_Port intensity; darker = higher urgency. Together these directly address Research Question 3: which regions present the highest priority for infrastructure expansion based on demand-supply gaps.

Modeling & Insights

Model used

A linear regression model (Ordinary Least Squares – OLS) was used to analyze the relationship between EV adoption and charging infrastructure across cities. The goal of this model is to quantify how changes in charging infrastructure availability influence the number of electric vehicles. In addition to the regression model, descriptive analytics techniques such as demand versus supply visualization, gap score analysis, utilization distribution, and geographic mapping were used to support and validate the findings. This combined approach ensures both statistical accuracy and practical business interpretation.

Variables (dependent & independent)

The model was constructed using EV count as the dependent variable and total charging ports as the independent variable. EV count represents the level of EV adoption in each city and serves as a measure of demand, while total charging ports, which includes both Level 2 and DC fast chargers, represents the availability of charging infrastructure and serves as a measure of supply. These variables were selected because they directly capture the core relationship between EV demand and infrastructure availability.

Key results

The regression results indicate a strong and statistically significant positive relationship between charging infrastructure and EV adoption. The model achieved an R-squared value of 0.879, which means that approximately 87.9% of the variation in EV counts across cities is explained by the number of charging ports. The coefficient for total charging ports is 21.07, indicating that each additional charging port is associated with an increase of approximately 21 EVs on average. The p-value is effectively zero, confirming that the relationship is statistically significant and not due to random variation. These results highlight the importance of charging infrastructure in supporting EV growth.

OLS Regression Results						
Dep. Variable:	ev_count	R-squared:	0.879			
Model:	OLS	Adj. R-squared:	0.878			
Method:	Least Squares	F-statistic:	3598.			
Date:	Tue, 28 Apr 2026	Prob (F-statistic):	9.68e-230			
Time:	14:47:04	Log-Likelihood:	-4056.5			
No. Observations:	499	AIC:	8117.			
Df Residuals:	497	BIC:	8125.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	169.5890	37.396	4.535	0.000	96.115	243.063
total_ports	21.0720	0.351	59.982	0.000	20.382	21.762
Omnibus:	420.402	Durbin-Watson:	1.819			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	16981.994			
Skew:	3.347	Prob(JB):	0.00			
Kurtosis:	30.784	Cond. No.	108.			

Figure 8: OLS Regression Results – EV Demand vs Charging Supply

Insights / findings

The regression findings are strongly supported by visual and descriptive analysis. As shown in Figure 9 (Demand vs Supply), there is a clear upward trend between charging ports and EV adoption, indicating that cities with more charging infrastructure tend to have higher EV counts. However, several cities lie above the general trend line, particularly those categorized as “No Infrastructure” and “Expansion Needed,” suggesting that EV demand in these areas exceeds the available charging capacity. This highlights the presence of significant demand–supply mismatches across multiple regions.

Further evidence of this imbalance is presented in Figure 10 (Top Underserved Cities), which identifies cities such as University Place, Duvall, South Hill, Camano Island, and Ferndale as having the highest gap scores. These cities have substantial EV adoption but little to no charging infrastructure, making them critical areas for expansion. The visualization clearly shows that even cities with some infrastructure, such as Mukilteo and Maple Valley, still experience significant gaps relative to their EV demand.

Additionally, Figure 11 (EV per Port Distribution) illustrates the uneven utilization of charging infrastructure across cities. The distribution is highly right-skewed, indicating that while many cities operate at moderate levels of demand per charging port, a subset experiences extremely high EV-to-port ratios. This suggests that charging infrastructure in these areas is under significant strain, potentially leading to congestion and reduced accessibility. At the same time, the presence of lower values indicates that some cities have relatively underutilized infrastructure.

Overall, these visualizations confirm that the primary challenge is not an overall shortage of charging stations, but rather an inefficient distribution of infrastructure across regions. Charging stations are concentrated in certain areas, while many growing cities remain underserved, reinforcing the need for more targeted and data-driven infrastructure expansion strategies.

Additionally, Figure 12 (Geographic Distribution of Charging Infrastructure) provides a spatial view of charging station locations across the region. The visualization shows that charging infrastructure is heavily concentrated in a few urban clusters, while many surrounding and peripheral areas have significantly fewer stations. This uneven geographic distribution reinforces the findings from the gap score and utilization analysis, highlighting that infrastructure is not lacking overall, but is inefficiently distributed across regions. As a result, several high-demand areas remain underserved, indicating clear opportunities for targeted infrastructure expansion.

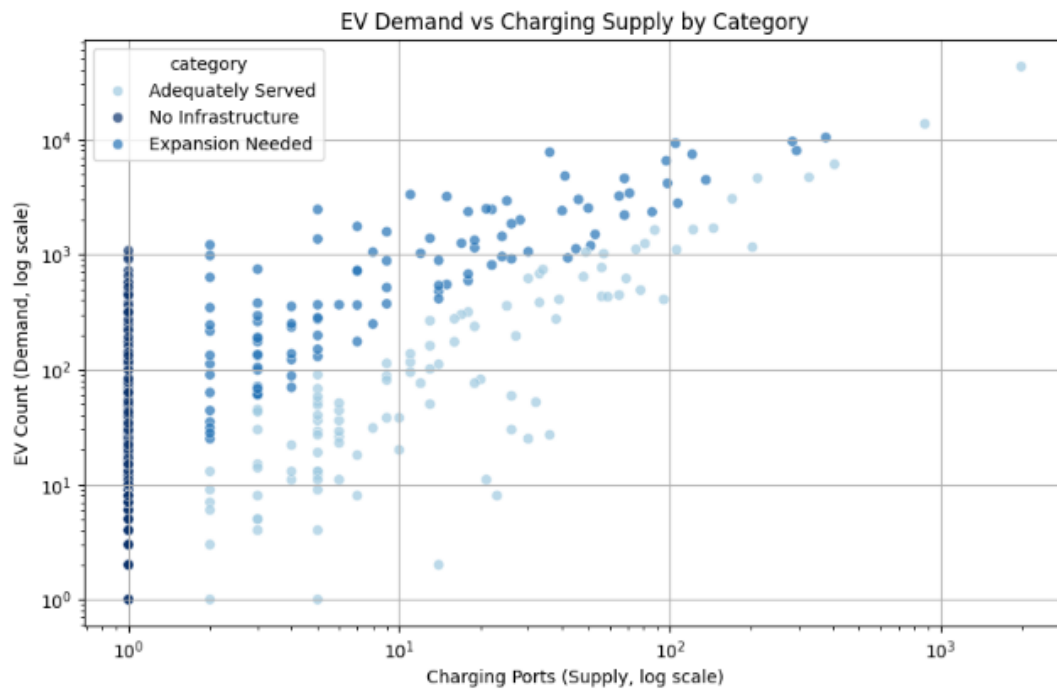


Figure 9: EV Demand vs Charging Supply (Log Scale)

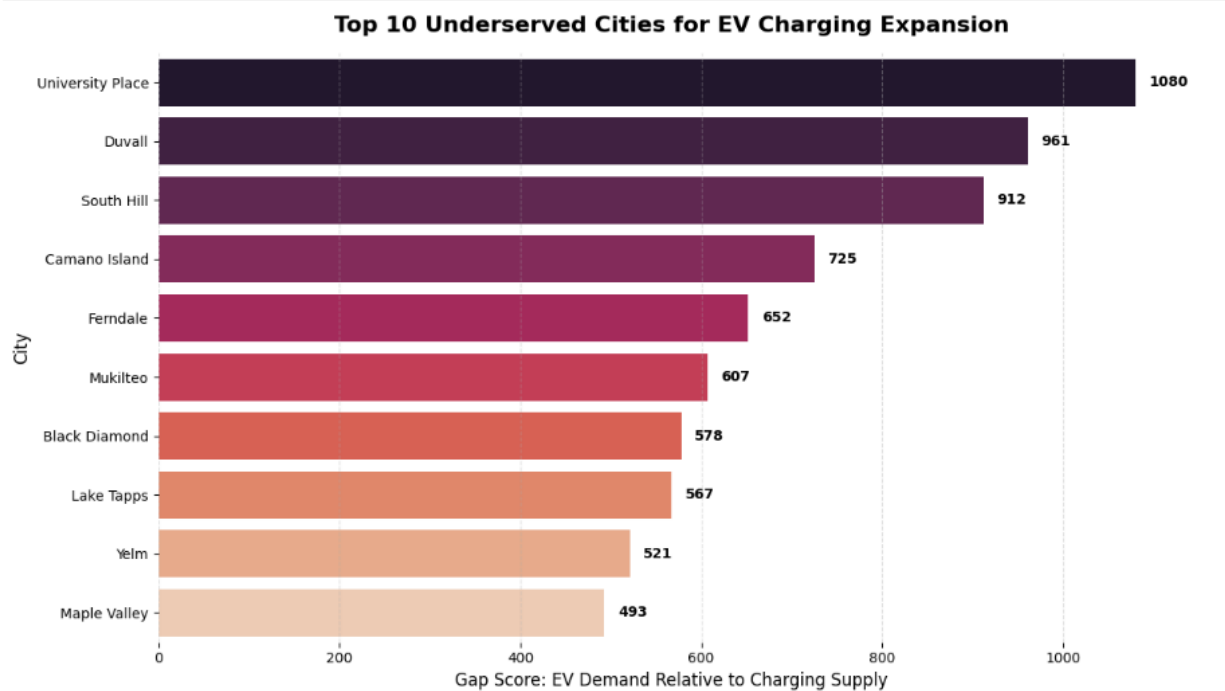


Figure 10: Top Underserved Cities

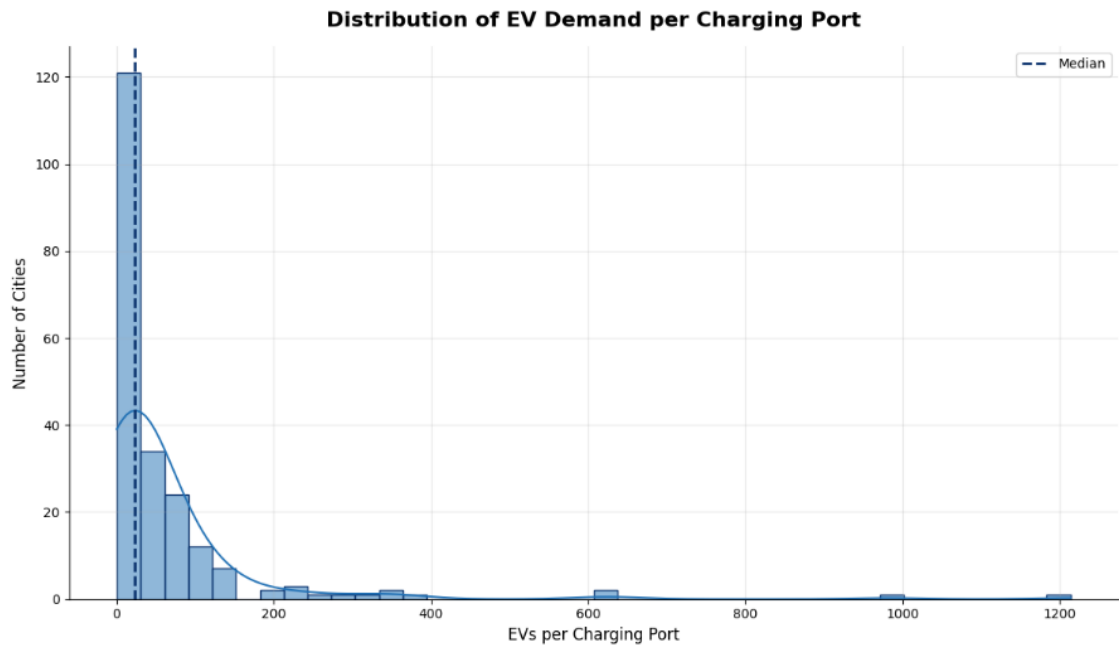


Figure 11: EV per Port Distribution

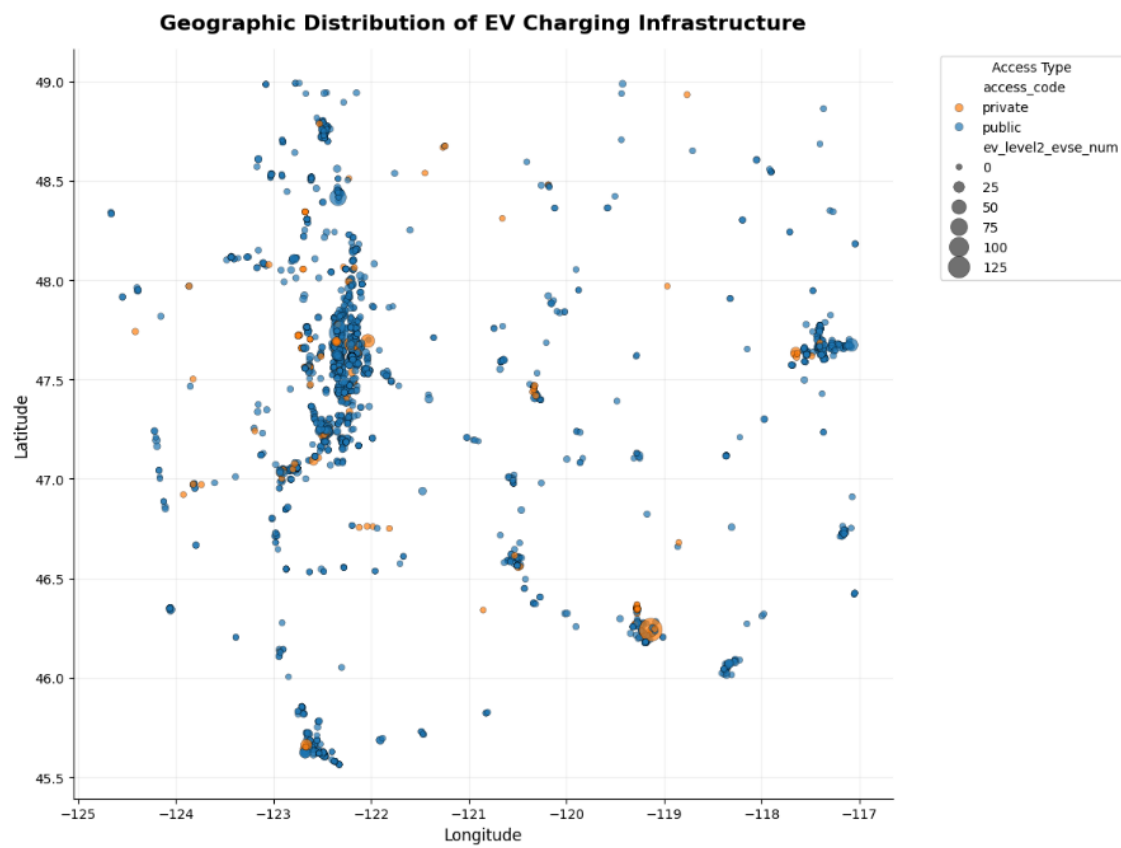


Figure 12: Geographic Distribution of Charging Infrastructure

Research Questions:

Which geographic regions exhibit the highest mismatch between EV adoption and charging station availability, indicating opportunities for infrastructure expansion?

The analysis reveals a significant mismatch between EV adoption and charging infrastructure across several cities in Washington State, particularly outside major urban centers. The demand vs. supply scatter plot shows a clear positive relationship between EV count and charging ports; however, many cities lie above the general trend line, indicating disproportionately high EV demand relative to available charging infrastructure. The distribution of EVs per charging port further supports this, showing a right-skewed pattern where most cities have manageable demand levels, but a subset of cities experience extremely high EV-to-port ratios, reflecting infrastructure strain.

The most critical gaps are identified through the gap score ranking, where cities such as University Place, Duvall, South Hill, Camano Island, and Ferndale exhibit the highest mismatch. These locations have substantial EV populations but little to no charging infrastructure, making them priority areas for expansion. Even cities like Mukilteo and Maple Valley, which have some charging stations, still show significant imbalance relative to their EV demand.

Additionally, the geographic distribution map highlights that charging infrastructure is heavily concentrated in urban clusters, while suburban and rural regions remain underserved. This uneven spatial distribution reinforces the finding that infrastructure development has not kept pace with EV adoption in several growing areas. Overall, the highest mismatch occurs in smaller and mid-sized cities with rising EV adoption but insufficient charging infrastructure, indicating clear opportunities for targeted investment to support future EV growth and ensure more equitable access to charging facilities.

How does charging infrastructure utilization vary across regions, and what are the implications for operational efficiency?

Charging infrastructure utilization varies significantly across cities, as estimated using the EV-per-port ratio, which serves as a proxy for demand pressure on available charging resources. The distribution of EVs per charging port is highly right-skewed, indicating that while many cities maintain relatively balanced utilization levels, a subset of cities experiences extremely high demand relative to available infrastructure. These high-utilization regions suggest that charging stations are operating under significant strain, leading to potential congestion, longer wait times, and reduced user convenience.

Conversely, some cities exhibit lower EV-per-port ratios, indicating relatively underutilized infrastructure. This imbalance suggests that inefficiencies in charging infrastructure are not due to an overall shortage but rather uneven distribution across regions. From an operational efficiency perspective, this highlights the need for better allocation of resources, ensuring that charging infrastructure is expanded in high-demand areas while avoiding overinvestment in regions with lower utilization.

Which regions present the highest priority for infrastructure expansion based on demand–supply gaps?

The highest priority regions for infrastructure expansion are identified using the gap score, which captures the mismatch between EV demand and available charging supply. Cities such as University Place, Duvall, South Hill, Camano Island, and Ferndale exhibit the most severe gaps, with high EV counts and little to no charging infrastructure. These locations represent critical bottlenecks where the lack of infrastructure may hinder further EV adoption.

The regression model further supports these findings by demonstrating a strong positive relationship between charging ports and EV adoption. This suggests that increasing charging infrastructure in underserved areas is likely to support continued growth in EV usage. Cities with high demand but insufficient supply therefore present strong opportunities for targeted investment, as expanding infrastructure in these regions can both alleviate current constraints and enable future adoption.

Recommendations & Conclusion

Recommendations

Based on the analysis conducted, our team recommends expanding charging infrastructure into cities with a high EV-per-port ratio. More specifically, ChargePoint should primarily focus on high ROI areas where EV demand is strong, but infrastructure is limited. Current areas that are underserved include University Place, Duvall, and South Hill; further suggestions can be seen in Figure 4. New analyses should be conducted yearly to ensure the company is not overbuilding into saturated markets. As expansion continues and these gaps are filled, ChargePoint could use infrastructure expansion strategically to drive future EV adoption.

Expected Impact

By focusing on areas with a high EV-per-port ratio, ChargePoint can maximize their profits at a new location. According to EVconnect.com, charging stations can earn between \$2,000 and \$50,000, depending on the charger's level, as shown in figure 13.

EV Charging Station Earnings Estimate Table		
Earnings Factor	Level 2 Charging	Level 3 (DC Fast) Charging
Average Earnings	\$2,000 – \$10,000 annually per unit	\$20,000 – \$50,000 annually per station
Revenue Potential	Steady income from workplaces, retail centers, and hospitality venues	High revenue potential at highways, transit hubs, and fleet depots
Utilization Impact	Profitable with moderate but consistent usage	Profitable with rapid turnover and high traffic flow
Customer Behavior	Encourages longer dwell time, leading to increased in-store spending	Attracts drivers seeking quick top-ups, boosting overall foot traffic
Revenue Models	Per kWh pricing, hourly rates, or subscription memberships	Premium pricing for speed and convenience, often per kWh or per session
Return on Investment	Moderate ROI over several years with steady usage	Faster ROI possible in high-demand locations with strong traffic flow

Figure 13: EV Charging Station Earnings Estimate Table (*Are EV Charging Stations Profitable?*, 2025)

Further examination should be carried out to determine which level of charger would be more profitable on a location-by-location basis. With this extra analysis, new infrastructure can be optimized for success.

Conclusion

ChargePoint's presence in Washington state will be far greater with the implementation of new charging infrastructure. Market analysis showed that there are plenty of areas where the EV adoption rate surpasses the number of charging stations. Data from the U.S. Department of Energy has been utilized to find the ratio of electric vehicles to charging stations, called the EV-per-port ratio. The EV-per-port ratio was used to discover underserved locations, including University Place, Duvall, and South Hill, where new charging infrastructure would be most beneficial.

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